HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Python** data pipelines for ETL processes, transforming raw data into structured formats for analytics and reporting on an enterprise ERP system.
- Engineered **Java Spring Boot** REST APIs to facilitate cross-system integrations, supporting data exchange between operational platforms and backend services.
- Managed production support for enterprise applications using **Splunk** and **Dynatrace**, responding to P1/P2 incidents and ensuring adherence to service level agreements.
- Designed and implemented **data pipelines** and **integrations** across asset management and operational systems, ensuring data quality, governance, and traceability.
- Led the development and evaluation of **Proof of Concepts** for asset intelligence solutions, defining success criteria and assessing emerging technologies like AI and Machine Learning.
- Collaborated with **stakeholders** across engineering and operations teams to understand asset management challenges and translate requirements into technical solutions for the iTRAK programme.
- Containerised **Java** applications using **Docker** and deployed them to **AWS EC2** instances, managing configurations and ensuring consistent operational environments.

SKILLS

Languages – Python, Java, SQL, R

Frameworks/Libraries – Spring Boot, pandas, NumPy, scikit-learn, Hibernate, JPA

Cloud/Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux

Data Engineering/BI – ETL, Data Modelling, Power BI, Tableau, Power Query, DAX, Data Warehousing

Tools/Platforms – Jenkins, Git, Jira, Confluence, Splunk, Dynatrace

Asset Management Concepts – ISO 55000 Principles, Condition Monitoring, Digital Twins

AI/ML Concepts – Machine Learning, Artificial Intelligence, Computer Vision, Advanced Analytics

PROJECTS

Asset Intelligence Data Pipeline for Infrastructure Monitoring

- Developed a **Python**-based data pipeline to ingest and process diverse asset data streams, including sensor readings and maintenance logs, from disparate sources.
- Implemented **SQL** transformations and data quality checks using **PostgreSQL**, ensuring integrity for downstream analytics and reporting on infrastructure health.
- Created **Power BI** dashboards to visualize key asset performance indicators and alert on anomalies, supporting proactive maintenance decisions for a critical infrastructure asset portfolio.

AI-Enabled Predictive Maintenance Model for Rolling Stock

- Engineered a **Python** feature extraction pipeline using **pandas** and **NumPy** to process historical maintenance records and operational data for rolling stock components.
- Trained a **scikit-learn** classification model to predict potential component failures, achieving approximately **0.84 AUC** on a dataset of 150,000 maintenance events.
- Developed a REST API using **FastAPI** to serve model predictions, enabling integration with an asset management system for early warning alerts.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)